

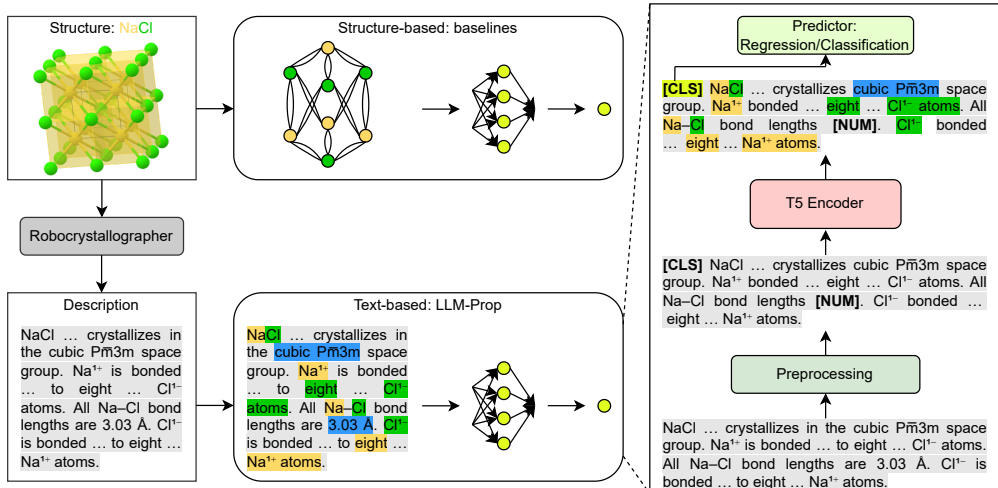


Figure 1: LLM-Prop architecture. On the left most of the figure, we show how we get the description from the crystal structure using Robocrystallographer. The middle part shows the comparison between our approach (text-based) and baselines (structure-based). The yellow colored information is related to Sodium (Na), green colored information is related to Chlorine (Cl), and blue colored information is related to additional information that text data provide such as space groups and bond distances. On the right most part of the figure, we then show our proposed LLM-Prop architecture (see 3.2 for more details).

focus was on classification tasks. Our work focuses on designing an efficient pipeline for the accurate prediction of crystal properties from text, for both regression and classification tasks. Our approach doesn’t rely on in-domain pretrained LLMs, has significantly fewer parameters, and can even play a crucial role in designing a more robust ranking and recommendation system for crystals because getting text embeddings from a good property predictor can improve performance on those tasks. Furthermore, we perform extensive and carefully-designed experiments, comparing performance against the strongest GNN-based approaches to shed light on the benefits of using text and highlight the shortcomings of current GNN-based models for crystals. Finally, we release to the public the curated dataset we used as a benchmark, called *TextEdge*, to accelerate NLP for materials science research.

3 LLM-Prop

3.1 Data Collection and Analysis

We collected the dataset used in this work from the Materials Project database (Jain et al., 2013) using the Materials Project free API² as of November 1, 2022. We focus on crystals with physical and electronic properties (band structures), which contain properties such as band gap, crystal volume, and an indicator of whether the band gap is direct or indirect. These properties have not yet been explored in other text-based methods and are usually hard to predict for GNN-based methods.

²<https://materialsproject.org/api>